

1. Introduction

Logickube¹ is a leading data and AI consultancy company that specializes in building scalable solutions in analytics, personalisation, cloud deployment and business intelligence. Our core team consists of senior data leaders that worked previously at WooliesX; the digital arm of Woolworths. With trillions and petabytes of data points generated by everyday Australians when they shop at Woolworths and BigW, our team are experts in big data analytics and modelling, building scalable personalisation and measurement products, and guiding stakeholders in a data and AI transformation journey.

Examples of AI techniques used by the team:

- Using graph neural networks to create a recipe personalisation engine
- Fitting Markov Chain models for user journey attribution
- Using gradient boosting machines to predict customer purchase propensity
- Building collaborative filter models for large scale recommender systems
- Applying statistical models (Poisson and Gamma) to estimate claim risks in motor and home insurance

To handle the large amount of data points provided by our clients, we are experts in cloud computing. We are an official partner of Google Cloud Platform and Databricks. Our experience has shaped us to be experts in handling petabyte-scale data and servicing personalisation and measurement in a high traffic environment. We deploy most of our work on Kubernetes Clusters that enable resources to be scaled up and down easily, containerisation of workloads for seamless deployment, and parallel processing of big data. We also utilize frameworks such as Apache Spark and Apache Beam to carry out data processing, model training and model serving.

In addition, we are experienced in productionising robotics solutions and have made remarkable contributions to the robotics society. We were the winner of Australia's first robotics dedicated accelerator, and we won the Best AI solution at SoftBank Robot World competition 2017, outperforming hundreds of other teams.

Our current clients are primarily retail / ecommerce giants, such as Woolworths and Endeavour Group. Throughout our engagement with them, we realized that retailers are moving into the world of AI to handle complex data-driven decisions quickly and accurately. Now is the time to harness data to digitally transform businesses to improve

¹ Website: <https://www.logickube.com/>

operational planning and forecasting, and interact with customers with a personalized user experience.

In particular, we provide extensive support to our clients in the digital world, i.e. eCommerce. We leverage customers' digital footprint data to enable and enhance informed decision making for the eCommerce business of our clients, which has been proven successful in the past. Such success, together with the growing interest in data-driven eCommerce solutions, motivated us to design this case study.

2. Business context

In the context of eCommerce, there are three critical concepts that you need to be familiar with:

- Customer journey
- Touch point
- Conversion

Customer journey is extensively mentioned and used in eCommerce. You may consider it as an end-to-end digital footprint that describes customer experience throughout different stages, from where they become aware of certain products till a purchase is completed.

A customer journey consists of one or more touch points, which are the 'actions' of the customer. Examples of touch points include, but are not limited to, webpage view (note that moving from one page to another is considered two separate touch points), clicks on link, add product to cart, complete a transaction, etc.

In reality, an observed customer journey does not necessarily cover all stages in eCommerce business. For example, a customer may become aware of a product, then do some browsing or research online via different channels, but end up not buying. Consequently, the concept of conversion indicates whether a customer journey is successful or not to the eCommerce business. Note that defining business success is rather subjective. For example, although completed transactions are widely used as business success, a marketing manager may consider 'customer open email and click on the promotion link' as business success, and those who satisfy this condition are considered 'converted' customers. In other words, conversion is directly related to how business success is measured, rather than anything else.

3. Data

For the case study, we used synthetic data, provided in the following two datasets:

- **customer_journey.csv**: details of customer journeys, including identifiers, touch points, and time to the end of the session.
- **results.csv**: the end result of each journey, namely, conversion or no conversion. You may think of a conversion as 'make a purchase'.

Refer to Appendix A1 and A2 for the description of data fields in each of the two datasets, respectively.

4. Your tasks

Based on the context and the datasets provided, you are required to develop a solution to each of the following business problems:

4.1. Predictive modelling

The first task is to better anticipate how customers will behave in the future, in terms of their engagement with the business. A common practice in this regard is predictive modelling, i.e. predicting a future state based on the observations in the past.

Use the data provided to develop the following TWO predictive models:

- Probability of a customer being converted on a **return** visit.
- At any given stage of a customer journey, predict how many steps it will take to reach a conversion.

In your solution, clearly state:

- Your algorithm(s)
- The dependent (target) variable
- The explanatory variables (model features), which can be either raw features from the data or engineered features which you create from the raw data. If you decide to use engineered features, clearly state how those are created.
- Assumptions you make
- How well your model performs, and how you evaluate the performance.

Tips:

- Creating model features (a.k.a. feature engineering) based on the information of past sessions plays a critical role in predictive modelling. It is recommended that you put in a decent amount of effort in this regard.
- Be mindful of 'information leakage'.
- Be aware that not all data records are suitable for modelling. For example, when predicting conversion on a return visit, only 2nd+ visits can be considered as you need at least one previous visit.

4.2. Stages of customers' life cycle

Understanding a customer's life cycle is of critical importance in eCommerce. Based on the data provided and your findings in Section 4.1, design a method to assign life cycle labels to customers according to their observed journeys, and verify the performance of your method.

Please note that the life cycle labels are subjective. You may define anything you like, as long as they make business sense and you are able to justify your decision.

Your set of labels must be mutually exclusive and collectively exhaustive. An example of such labels is early stage / mid stage / late stage / no conversion.

4.3. Business success attribution

For a given amount of business success, an attribution method aims to determine how much 'credit' each contributing factor deserves. As a simple example, if email and mobile apps are used in a promotion campaign with $x\%$ and $y\%$ contribution, respectively, then we need to determine the value of x and y , subject to $x + y = 100$.

For the sake of simplicity, we define business success as conversion.

4.3.1. Contribution of each channel

To optimise marketing efficiency, it is always important to understand the contribution of each channel.

Using the data provided, perform the following two tasks:

- Calculate the overall conversion rate.

- Attribute the conversions to the eight channels, namely, Organic, Direct, Email Marketing, Google Search, Google Display, Youtube, Facebook, and Instagram, as shown in the '*channel*' data field. You may decide how the attribution results are reported. An example is to calculate eight weights, corresponding to the eight channels respectively, which sum up to 100%.

4.3.2. Impact of journey lengths on attribution

In reality, customer journeys may be of various lengths. As shown in the data, some journeys only had a single touch point, whereas some others had more than 20. The business is keen to find out whether the lengths of journeys would have any impact on the attribution results.

Using the data provided, answer the following two questions:

- Are the attribution results from shorter journeys comparable to those from longer journeys? Note that it is up to you to design a solution to this problem and there are several ways this can be investigated:
 - You may separate the journeys into groups by their lengths (e.g. less than 3 touch points, 3-5 touch points, 6-10 touch points, and 10+ touch points), perform attribution on each group, and compare the results
 - Alternatively, you may want to conduct your analysis in an incremental manner, i.e., perform attribution on single touch point journeys, and then on journeys with length ≤ 2 , and then on journeys with length ≤ 3 , and so on.
 - Another option may be to incorporate the duration of time each journey takes, i.e., separate journeys into groups of similar life cycle duration and assign attribution to each group, and compare the results
- Would the attribution results differ if the journeys are 'collapsed'? 'Collapse' means merging two or more same touch points that are adjacent. For example, the journey 'Direct > Facebook > Facebook > Instagram' collapses to 'Direct > Facebook > Instagram'.

4.3.3. Attribution on different levels of granularity

As you may have noticed, the eight channels naturally form three groups as follows:

- Free channels - Organic, Direct, Email Marketing
- Google paid channels - Google Search, Google Display, Youtube
- Meta paid channels - Facebook, Instagram

The business wants to understand whether the attribution results remain consistent across the following two levels of granularity:

- *Level 1* - the three groups of channels, i.e., free channels, Google paid channels, and Meta paid channels
- *Level 2* - the eight channels that you worked with in the previous sections.

Using the data provided and the results you obtained in the previous sections, answer the following two questions:

- Do the attribution results at Level 2 add up to the results at Level 1?
- If not, is there any method that guarantees consistency across different levels of granularity? That is, no matter what data you deal with, the attribution results at a more granular level always add up to those at a more aggregated level.

You are encouraged to either investigate existing methods or develop your own approach to attribution. Clearly state the assumptions and limitations of your approach, if any, if you choose the latter.

Appendix

A1. Description of data fields in *customer_journey.csv*

Field name	Type	Description
user_id	INTEGER	Unique user IDs. Nominal data.
session_id	INTEGER	ID of the session(s) that each user had. Ordinal data, in the sense that larger value means the session was more recent.
order	INTEGER	Integers denoting the sequence of touch points (channels) in a journey.
channel	STRING	eCommerce channels that are touch points of user journeys.
time_till_session_end	FLOAT	Time (in seconds) lapsed from a touch point to the end of a session.

A2. Description of data fields in *results.csv*

Field name	Type	Description
user_id	INTEGER	Unique user IDs. Nominal data.
session_id	INTEGER	ID of the session(s) that each user had. Ordinal data, in the sense that larger value means the session was more recent.
conversion	INTEGER	A flag indicating whether the corresponding user journey led to a conversion. 1 means converted while 0 means not converted.